

Documentación de la práctica: Sonido

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Asignatura: Periféricos y dispositivos de interfaz humana

Universidad: UGR

1. Introducción

En esta práctica se ha trabajado en RStudio para realizar la práctica de edición y visualización de audio.

2. Herramientas utilizadas

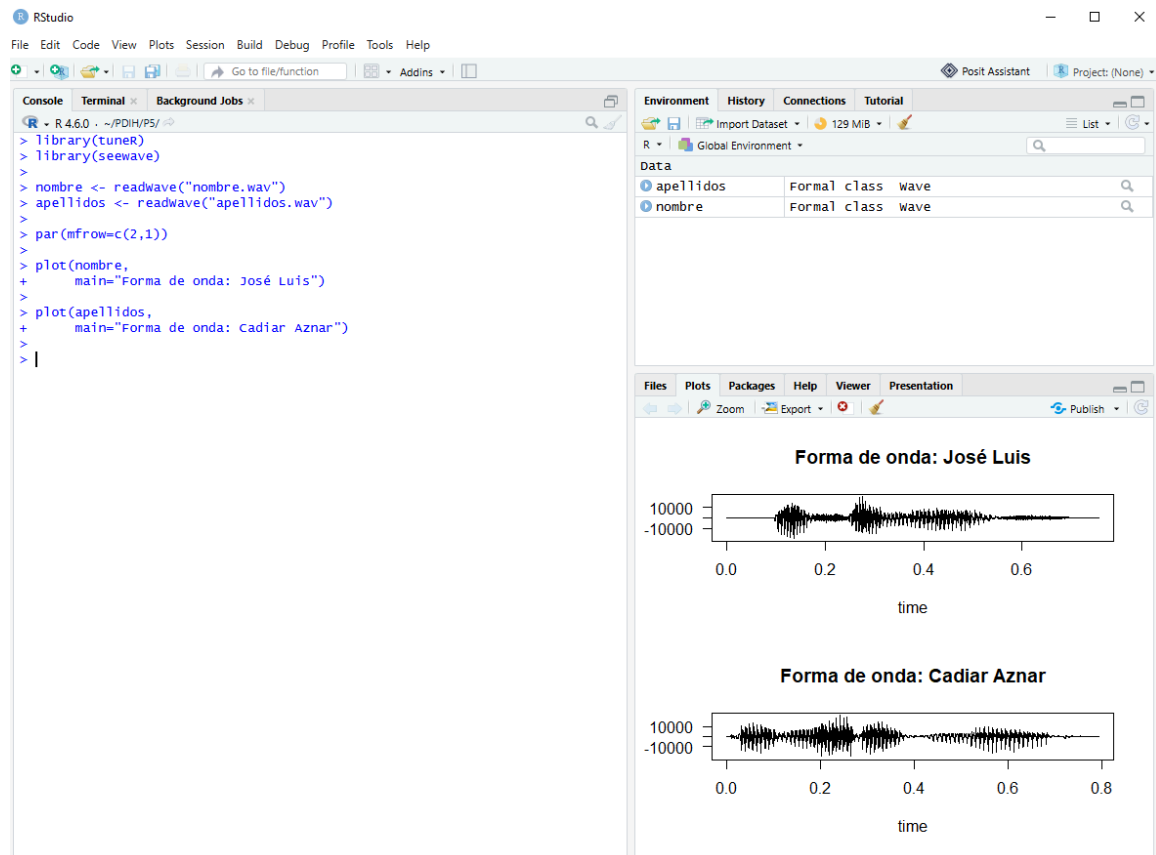
R como lenguaje para la realización de la práctica

RStudio para la edición y visualización de audio.

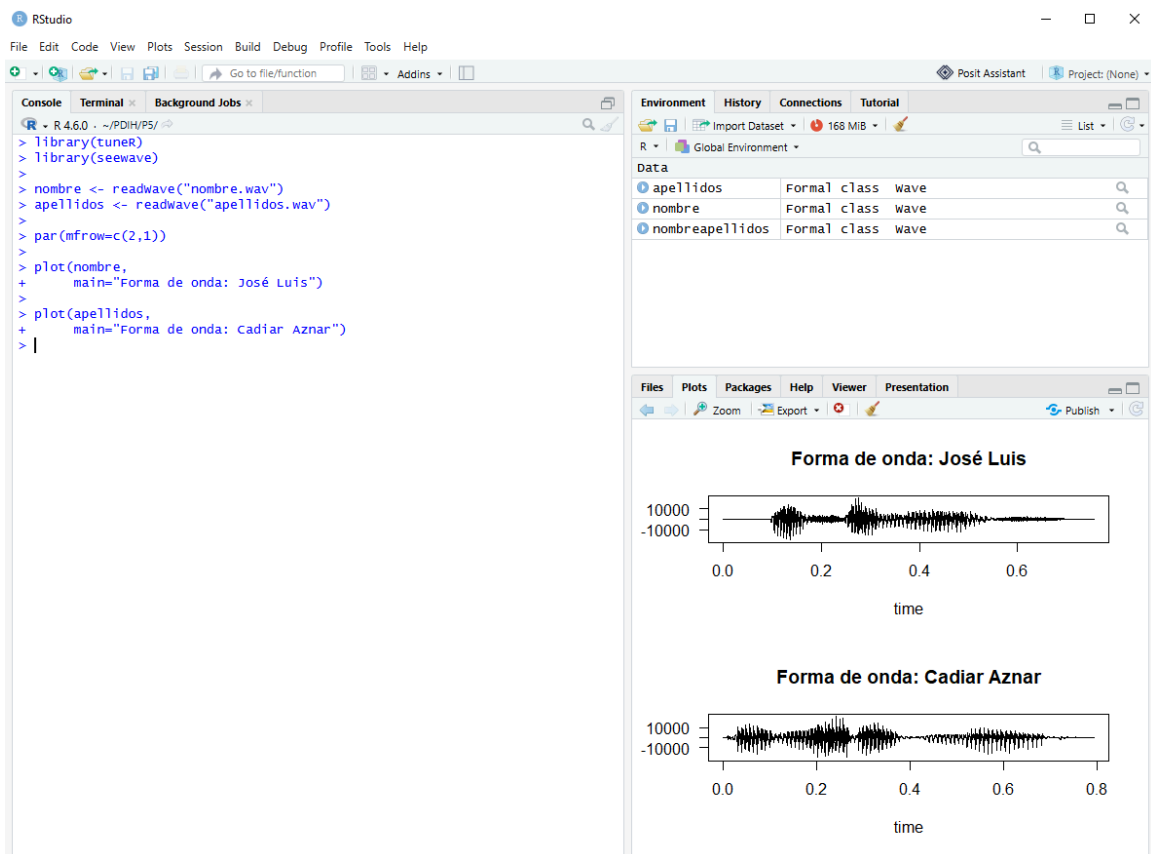
3. Realización de los ejercicios

He hecho el desarrollo de los ejercicios utilizando el programa RStudio, los comandos básicos aportados en el guión de la práctica.

4. Ejercicio 1



5. Ejercicio 2



6. Ejercicio 3

```
> summary(nombre)
```

```
Wave object
```

```
Number of Samples: 36335
Duration (seconds): 0.76
Samplingrate (Hertz): 48000
Channels (Mono/Stereo): Mono
PCM (integer format): TRUE
Bit (8/16/24/32/64): 16
```

```
Summary statistics for channel(s):
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-19114.0	-542.0	0.0	-16.6	871.0	20208.0

```
>
```

```
> summary(apellidos)
```

```
Wave object
```

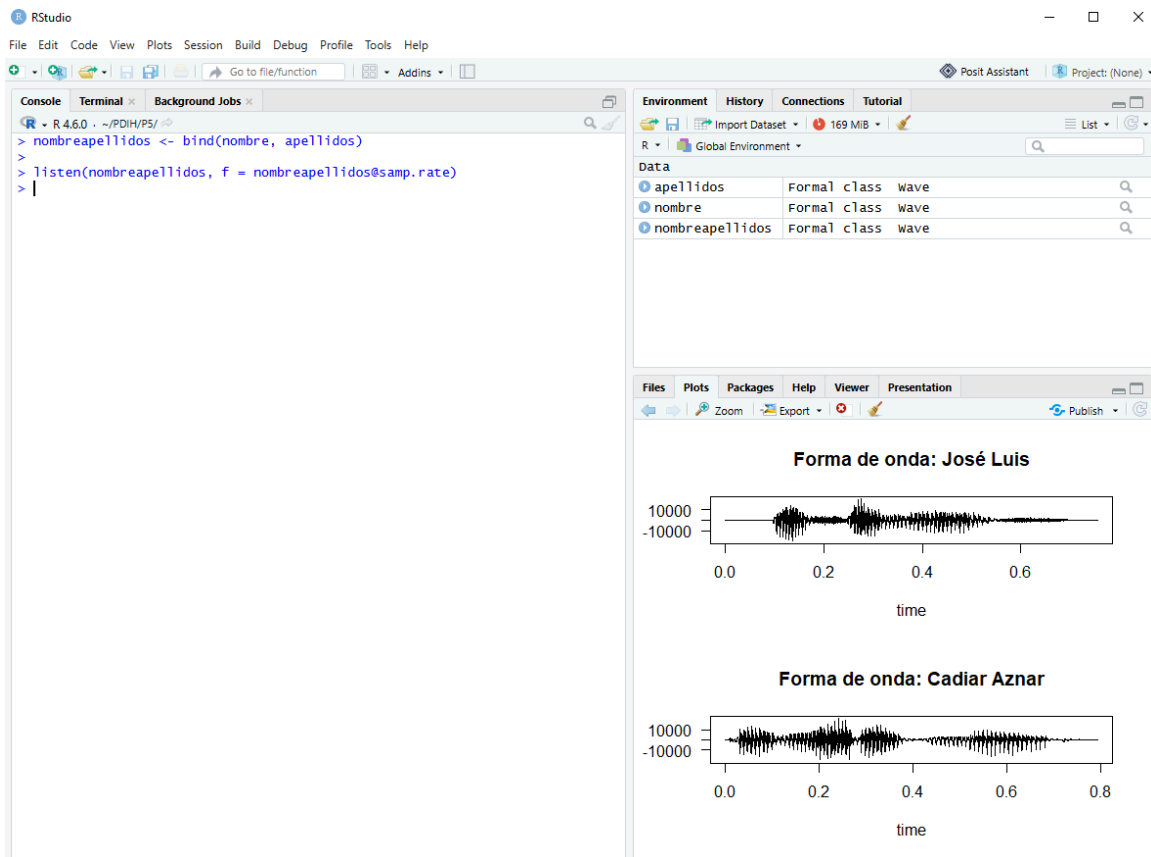
```
Number of Samples: 38059
Duration (seconds): 0.79
Samplingrate (Hertz): 48000
Channels (Mono/Stereo): Mono
PCM (integer format): TRUE
Bit (8/16/24/32/64): 16
```

```
Summary statistics for channel(s):
```

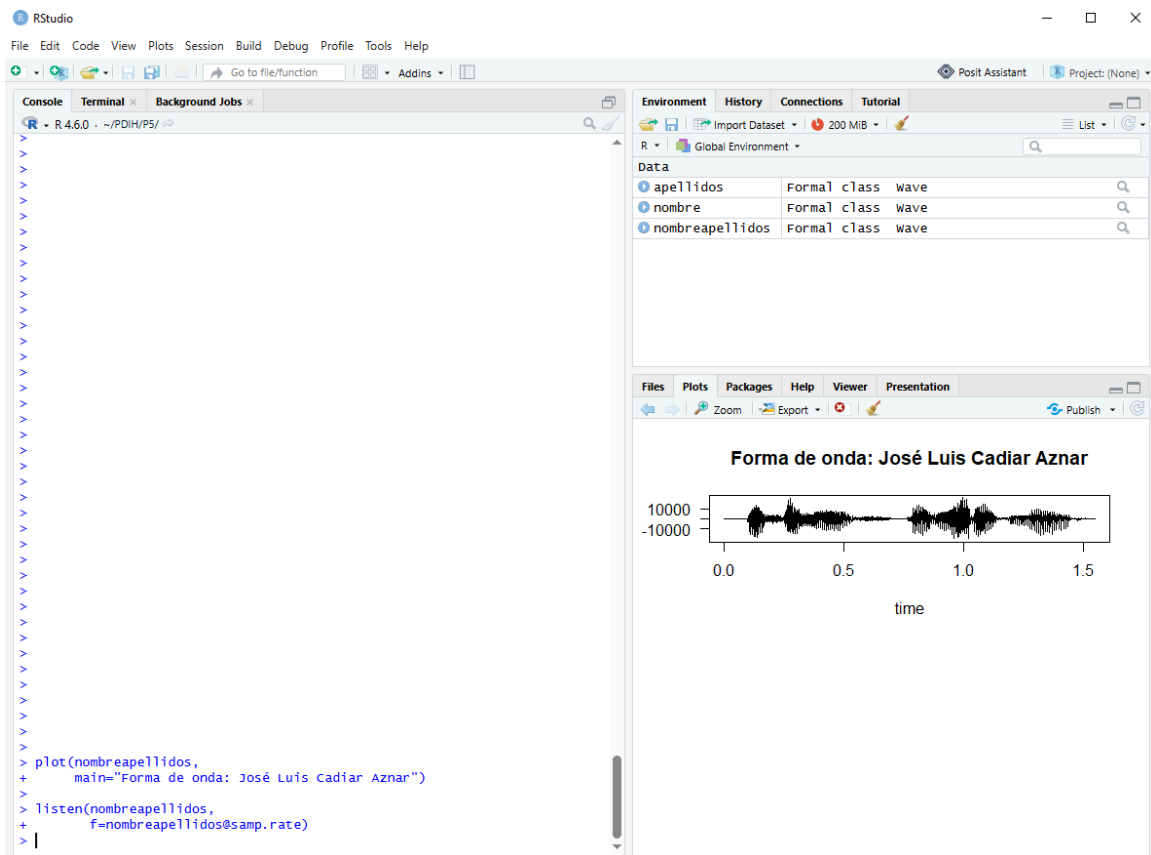
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-20051.00	-1115.50	159.00	-24.69	1596.00	21672.00

```
>
```

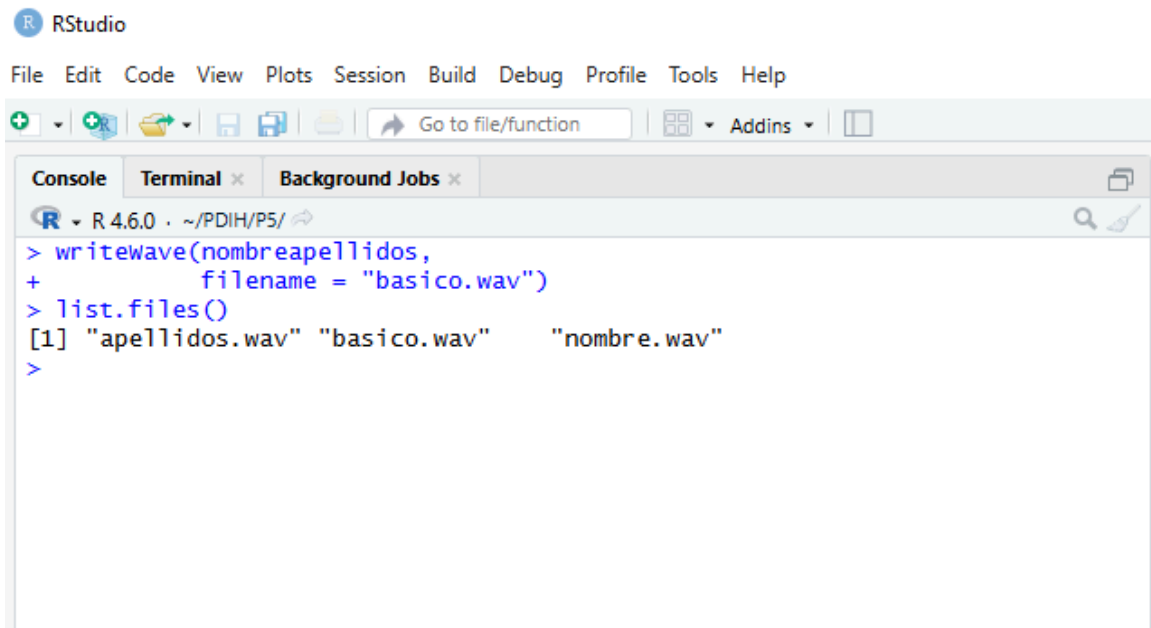
7. Ejercicio 4



8. Ejercicio 5



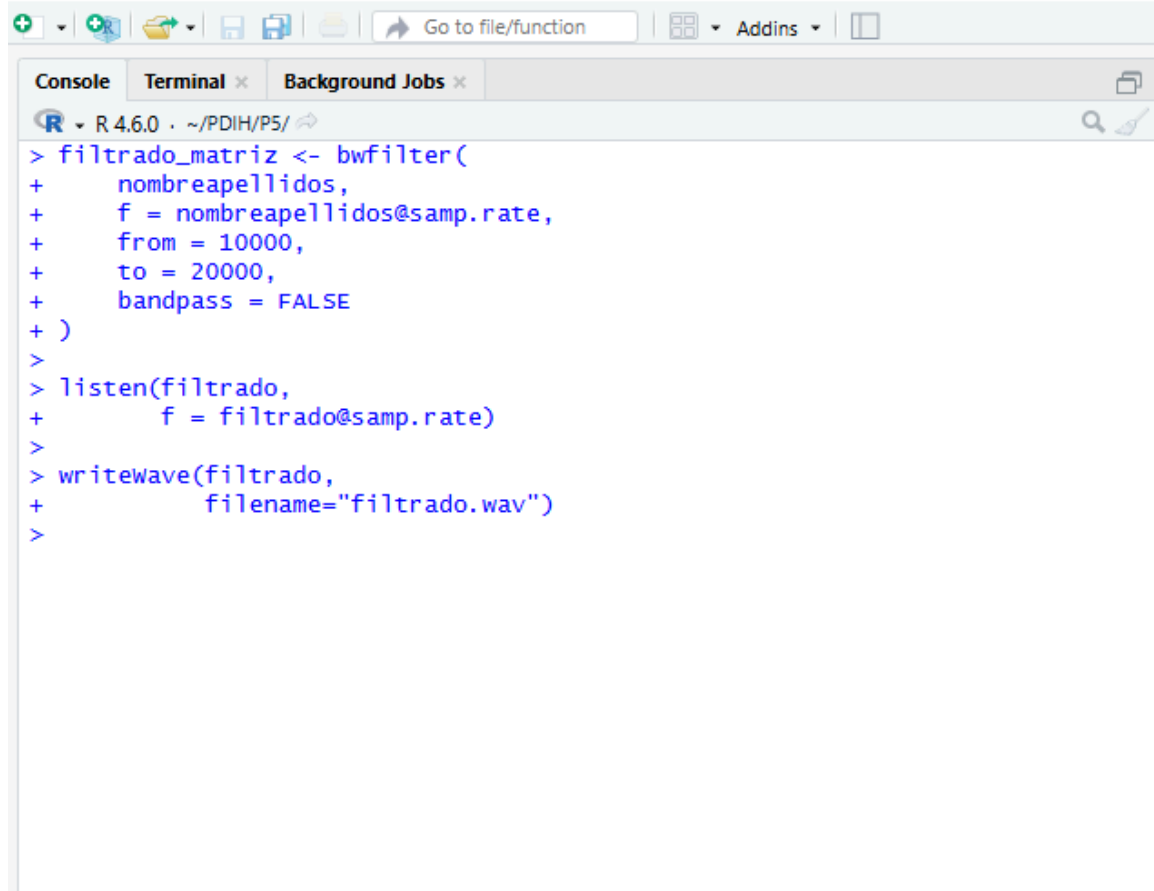
9. Ejercicio 6



10. Ejercicio 7

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

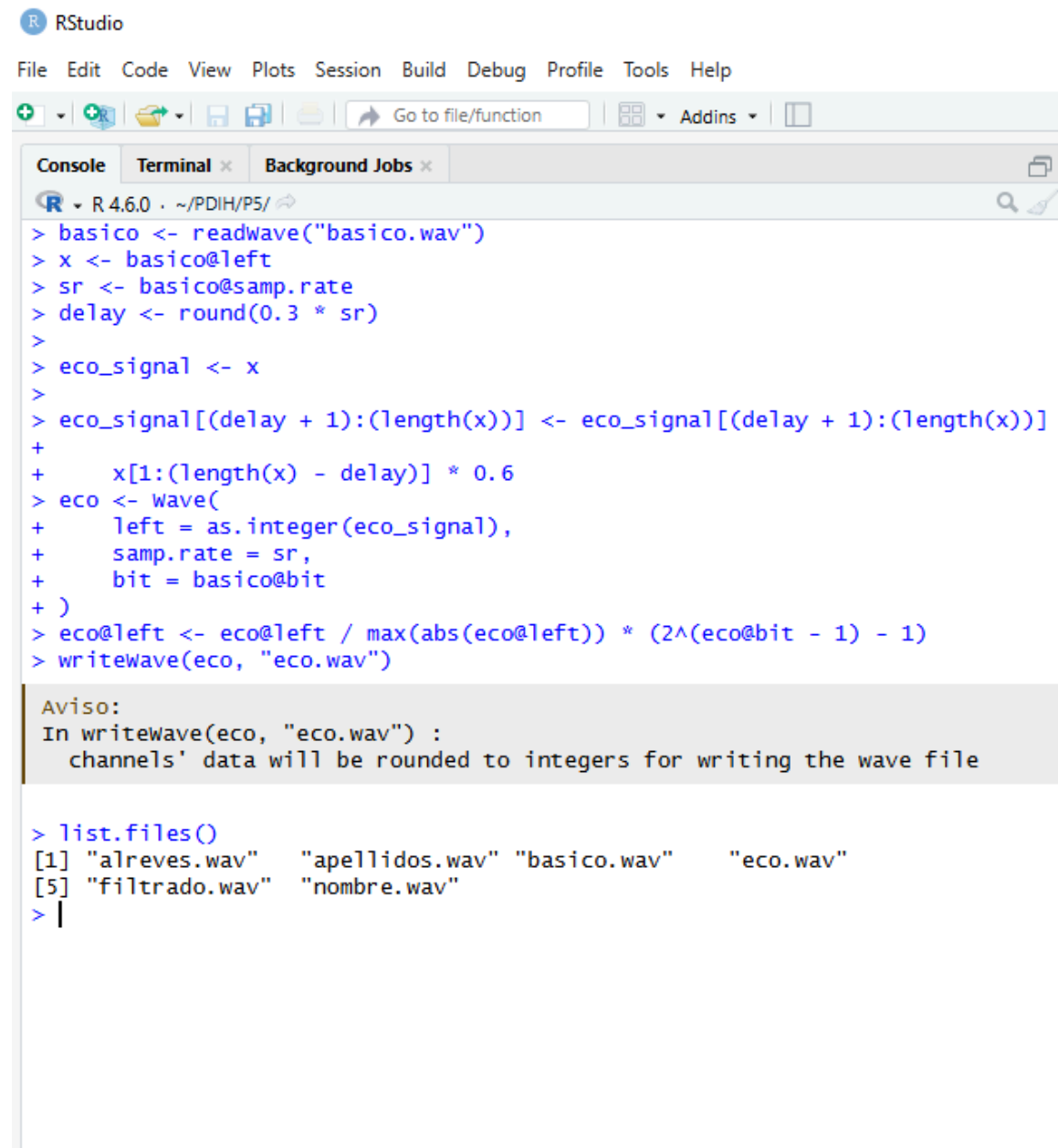


The screenshot shows the RStudio interface with the console pane active. The console displays the following R code:

```
> filtrado_matriz <- bwfilter(  
+   nombreapellidos,  
+   f = nombreapellidos@samp.rate,  
+   from = 10000,  
+   to = 20000,  
+   bandpass = FALSE  
+ )  
>  
> listen(filtrado,  
+   f = filtrado@samp.rate)  
>  
> writewave(filtrado,  
+   filename="filtrado.wav")  
>
```

The console title bar indicates the R version is 4.6.0 and the working directory is ~/PDIH/P5/. The interface also shows tabs for Terminal and Background Jobs, and a toolbar with icons for file operations and navigation.

11. Ejercicio 8



The screenshot shows the RStudio interface with the console pane active. The console displays the following R code and its output:

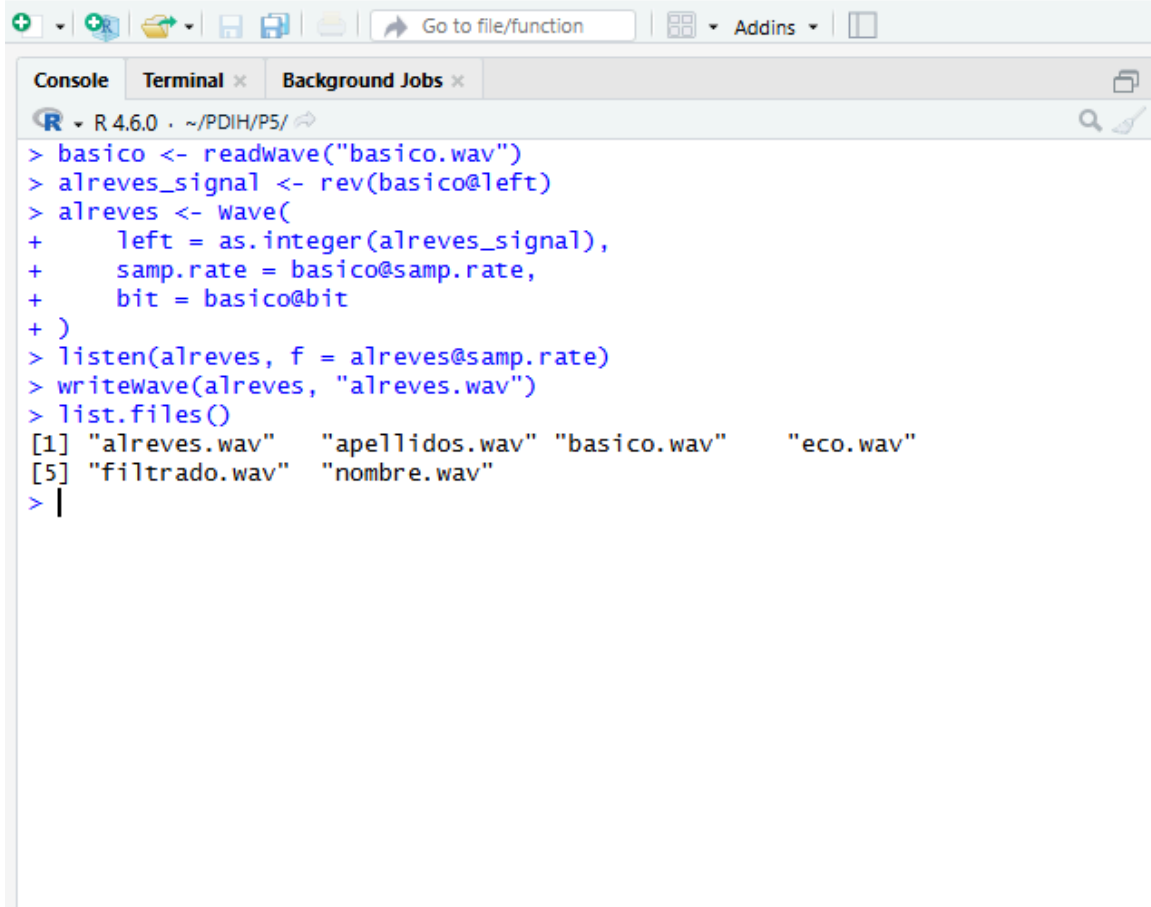
```
> basico <- readwave("basico.wav")
> x <- basico@left
> sr <- basico@samp.rate
> delay <- round(0.3 * sr)
>
> eco_signal <- x
>
> eco_signal[(delay + 1):(length(x))] <- eco_signal[(delay + 1):(length(x))]
+   + x[1:(length(x) - delay)] * 0.6
> eco <- wave(
+   left = as.integer(eco_signal),
+   samp.rate = sr,
+   bit = basico@bit
+ )
> eco@left <- eco@left / max(abs(eco@left)) * (2^(eco@bit - 1) - 1)
> writewave(eco, "eco.wav")
```

A warning message is displayed in a grey box:

```
Aviso:
In writewave(eco, "eco.wav") :
  channels' data will be rounded to integers for writing the wave file
```

The console continues with the following code and output:

```
> list.files()
[1] "alreves.wav"  "apellidos.wav" "basico.wav"    "eco.wav"
[5] "filtrado.wav" "nombre.wav"
> |
```

The image shows the RStudio application window. At the top is the menu bar with options: File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu bar is a toolbar with icons for file operations (new, open, save, etc.) and a search bar. The main workspace area is divided into three tabs: Console, Terminal, and Background Jobs. The Console tab is active, showing the R prompt and the following code:

```
> basico <- readwave("basico.wav")
> alreves_signal <- rev(basico@left)
> alreves <- wave(
+   left = as.integer(alreves_signal),
+   samp.rate = basico@samp.rate,
+   bit = basico@bit
+ )
> listen(alreves, f = alreves@samp.rate)
> writewave(alreves, "alreves.wav")
> list.files()
[1] "alreves.wav"    "apellidos.wav"  "basico.wav"     "eco.wav"
[5] "filtrado.wav"   "nombre.wav"
```

The code reads a WAV file named "basico.wav", reverses the left channel, and writes the result to "alreves.wav". It then lists the files in the current directory, showing the output of the `list.files()` function.